

# MATH3405 – Differential Geometry

## Semester 2, 2025

### Assignment 2

To be submitted before Friday 31 October at 5pm.

Total number of marks = 80. Optional tasks will not be taken into account for grading and may be skipped.

**Q 1 [30 marks] (Surfaces of revolution)** Let  $\alpha : I \rightarrow \mathbb{R}^2$ ,  $\alpha(t) = (\varphi(t), \psi(t))$ , be a regular curve without self-intersections, with  $\varphi(t) > 0$  for all  $t \in I$  (here  $I \subset \mathbb{R}$  is an open interval). **Furthermore, assume that  $\psi'(t) > 0$  for all  $t \in I$ .** We now imagine the curve  $\alpha$  as sitting inside the  $yz$ -plane in  $\mathbb{R}^3$ , and rotate it around the  $z$ -axis. The surface thus obtained, denoted by  $S$ , is called a *surface of revolution*. More precisely,

$$S = \{(\varphi(v) \cos u, \varphi(v) \sin u, \psi(v)) : u \in [0, 2\pi), v \in I\}.$$

- (a) Prove that  $S$  is a regular surface.
- (b) Show that  $S$  is diffeomorphic to the cylinder

$$C := \{(\cos u, \sin u, v) : u \in [0, 2\pi), v \in I\}.$$

- (c) Assuming that  $\alpha$  is parametrised by arc-length parameter, compute the coefficients of the first fundamental form in some parametrisation of  $S$ .

**Q 2 [50 marks] (Geodesics on surfaces of revolution)** Let  $S$  be a surface of revolution and consider for it the parametrisation

$$\mathbf{x}(u, v) = (\varphi(v) \cos u, \varphi(v) \sin u, \psi(v)).$$

Assume in addition that the condition  $(\frac{d\varphi}{dv})^2 + (\frac{d\psi}{dv})^2 \equiv 1$  holds.

**2a** Prove that a curve  $\alpha(t) = \mathbf{x}(u(t), v(t))$  is a geodesic of  $S$  if and only if it satisfies

$$\ddot{u} + 2 \frac{d\varphi}{dv} \frac{1}{\varphi} \dot{u} \dot{v} = 0, \quad \ddot{v} - \frac{d\varphi}{dv} \varphi (\dot{u})^2 = 0,$$

where here and in what follows the dot denotes derivative with respect to  $t$ .

**[10 marks]**

**2b** Prove that meridians ( $u \equiv u_0$  constant) are geodesics. Prove that a parallel  $v \equiv v_0$  is a geodesic if and only if  $v_0$  is a critical point for  $\varphi(v)$ .

**[10 marks]**

- 2c (Clairaut's relation)** Let  $\alpha(t)$  be a geodesic and denote by  $\theta(t)$  the angle between  $\dot{\alpha}(t)$  and the corresponding parallel. Show that

$$\varphi(v(t)) \cos \theta(t) \equiv c$$

is constant along the geodesic. (Notice that this statement is independent of the parametrisation  $\mathbf{x}$ , since  $\varphi(v)$  here represents the distance between a point  $\mathbf{x}(u, v)$  and the axis of revolution. ) [10 marks]

- 2d** Conversely, show that if Clairaut's relation is satisfied along a curve  $\alpha : I \rightarrow S$  on a surface of revolution, and there is no non-empty open interval  $J \subset I$  such that  $\alpha(J)$  is contained in a parallel, then  $\alpha$  is a geodesic. [10 marks]

- 2e** The *pseudosphere* is the surface of revolution parametrized by

$$\mathbf{x}(u, v) = \left( \frac{1}{v} \cos u, \frac{1}{v} \sin u, \sqrt{1 - \frac{1}{v^2}} - \cosh^{-1} v \right), \quad u \in \mathbb{R}, v > 1.$$

It has constant Gauss curvature equal to  $-1$ . Use the above items to compute the geodesics of the pseudosphere. (It is enough to express them as images  $\mathbf{x}(u(t), v(t))$  of curves  $(u(t), v(t))$  in the  $(u, v)$ -plane  $\mathbb{R}^2$ ; the latter can be presented in implicit form.) [10 marks]

- 2f Optional:** Prove that any geodesic of the paraboloid of revolution  $z = x^2 + y^2$  which is not a meridian intersects itself an infinite number of times.

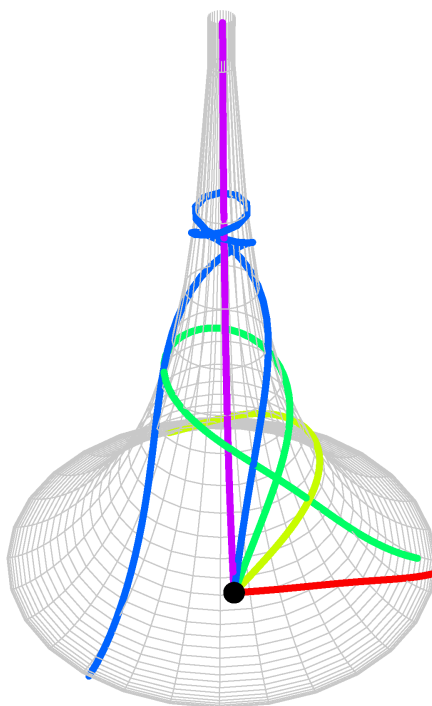


Figure 1: Geodesics on the pseudosphere

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<sup>1</sup>Image taken from the book Gray, A.; Abbena, E.; Salamon, S., *Modern differential geometry of curves and surfaces with Mathematica* ®. Third edition. Studies in Advanced Mathematics. Chapman & Hall/CRC, Boca Raton, FL, 2006. xxiv+984 pp.